MECHATRONICS AND IOT

ASSIGNMENT REPORT - 7

TITLE:

Design and simulate Smart Elevator Automation and Human Detection System using PIR Sensor. The system automatically detects human presence and controls elevator operations for improved safety and automation.

DONE BY ,

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Project Overview:

The Smart Elevator Automation and Human Detection System is an embedded automation project designed to improve the safety, efficiency, and intelligent operation of elevators using a PIR (Passive Infrared) sensor. The system detects the presence of a person near or inside the elevator and automatically controls elevator operations based on the detected human activity.

The PIR sensor detects infrared radiation emitted by the human body and sends a signal to a microcontroller such as an ESP32 or Arduino. Based on this signal, the controller determines whether a person is present and initiates the required elevator operation. The system can control functions such as door opening/closing, elevator movement, floor selection, and automatic stopping.

A key safety feature is human detection during door operation. If a person is detected while the elevator doors are closing, the controller can stop or reopen the doors, reducing the possibility of passengers being trapped or injured. The system can also prevent unnecessary elevator operation when no person is detected, thereby reducing energy consumption.

The proposed system can be designed and simulated using platforms such as Arduino IDE, Proteus, or Tinkercad, depending on the required level of simulation. A motor or motor driver can be used to represent elevator movement, while LEDs or a display can indicate the current floor and operating status.

Problem Statement:

Conventional elevators generally depend on manual button operation and basic control systems, which may not effectively detect human presence during critical operations such as door closing. This can lead to unnecessary elevator movement, increased energy consumption, and potential safety risks, especially when a person is entering or exiting the elevator.

Therefore, there is a need for an automated and intelligent elevator system capable of detecting human presence and responding appropriately. The proposed project uses a PIR sensor and microcontroller to detect people and automatically control elevator operations such as door opening, closing, and movement.

The system aims to improve passenger safety, operational efficiency, and automation while providing a simple and cost-effective solution suitable for smart-building applications.

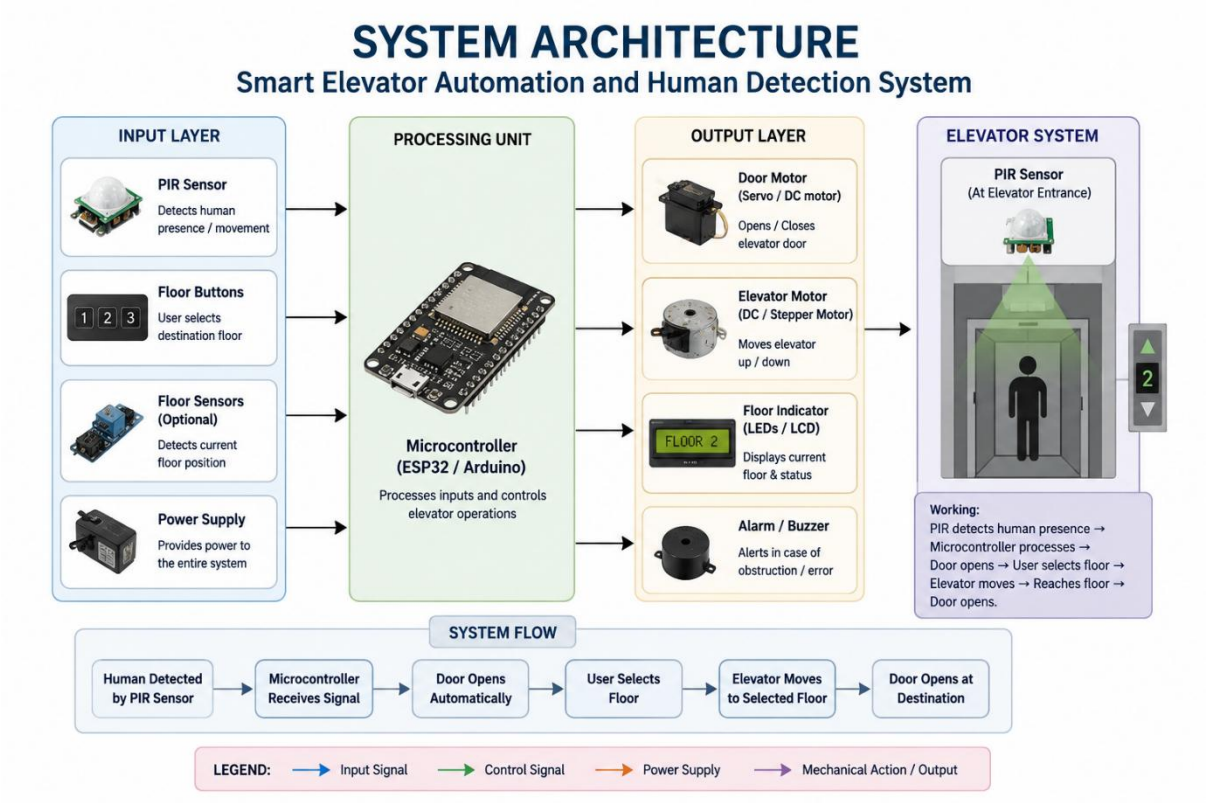
Proposed Solution:

The proposed system is a Smart Elevator Automation and Human Detection System that uses a PIR sensor and microcontroller to detect human presence and automatically control elevator operations.

The PIR sensor detects the movement/presence of a person near the elevator and sends a signal to the microcontroller. Based on this input, the controller operates the elevator door and motor mechanism. When a person is detected, the elevator can automatically activate and open the door. If no person is detected for a predefined period, the system can close the door and enter an idle state.

As a safety feature, if the PIR sensor detects a person while the door is closing, the microcontroller immediately stops the closing operation and reopens the door. The elevator motor can also be controlled according to the required floor, with LEDs or an LCD used to indicate the elevator's status.

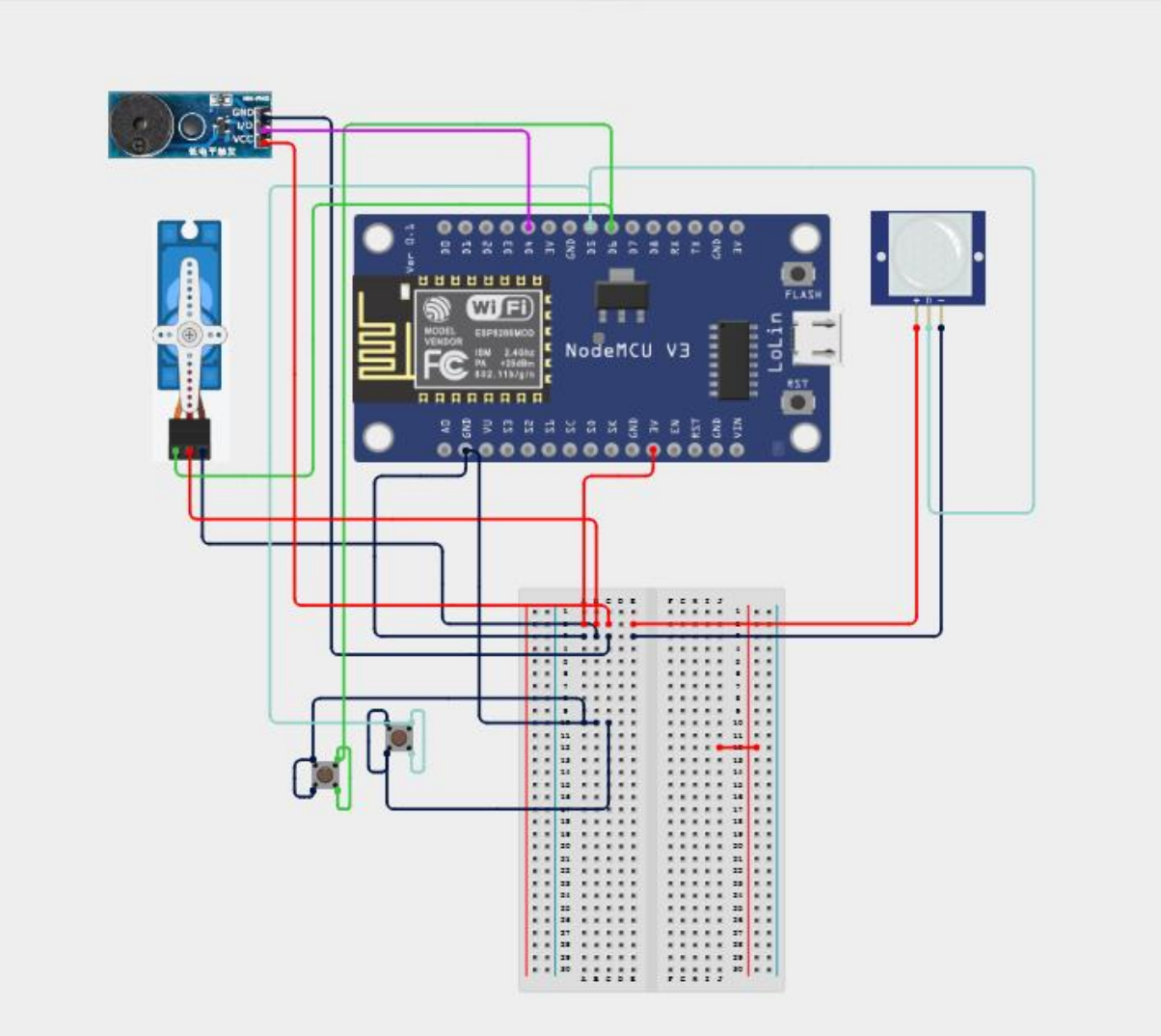
System Architecture:



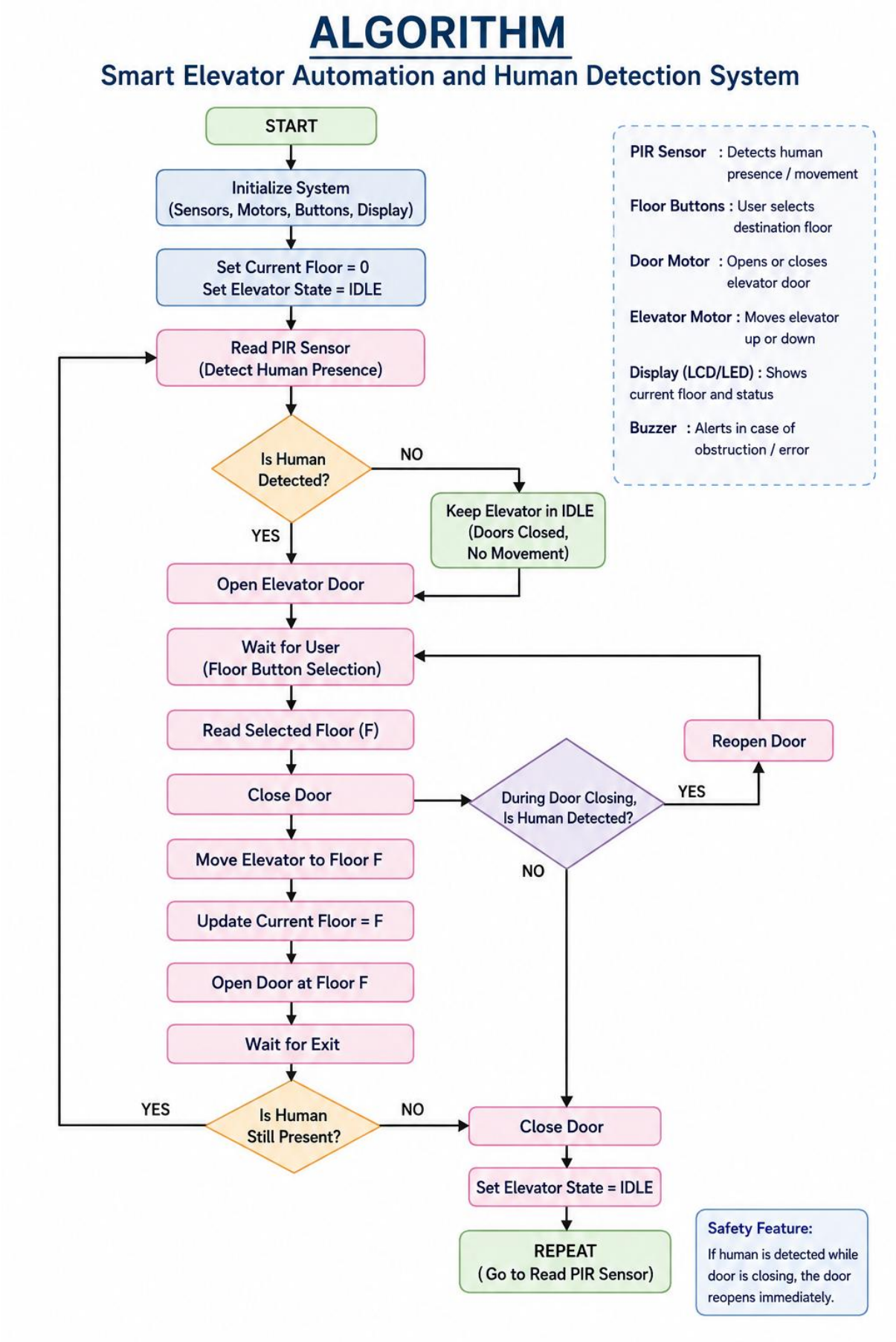
Circuit Table :

Component	Pin/Terminal	ESP8266 Connection	Purpose
PIR Sensor	VCC	3.3V	Power supply
Servo Motor	VCC	5V	Power supply
LCD Display	VCC	3.3V	Safe indication
ESP8266	VIN/USB	5V USB	Hazardous indication
Buzzer	Positive (+)	GPIO 14	Alert

Circuit Design:



Algorithm:


ALGORITHM
Smart Elevator Automation and Human Detection System
START
Initialize System
(Sensors, Motors, Buttons, Display)
Set Current Floor = 0
Set Elevator State = IDLE
Read PIR Sensor
(Detect Human Presence)
Is Human Detected?
NO
YES
Keep Elevator in IDLE
(Doors Closed,
No Movement)
Open Elevator Door
Wait for User
(Floor Button Selection)
Read Selected Floor (F)
Close Door
During Door Closing,
Is Human Detected?
YES
Reopen Door
NO
Move Elevator to Floor F
Update Current Floor = F
Open Door at Floor F
Wait for Exit
YES
Is Human
Still Present?
NO
Close Door
Set Elevator State = IDLE
REPEAT
(Go to Read PIR Sensor)
PIR Sensor : Detects human presence / movement
Floor Buttons : User selects destination floor
Door Motor : Opens or closes elevator door
Elevator Motor : Moves elevator up or down
Display (LCD/LED) : Shows current floor and status
Buzzer : Alerts in case of obstruction / error
Safety Feature:
If human is detected while door is closing, the door reopens immediately.

Coding:

```
#include <ESP8266WiFi.h>
#include <Servo.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
// ============================================================
// WIFI SETTINGS
// ============================================================
#define WIFI_SSID       "Xiaomi 11i"
#define WIFI_PASSWORD   "12345678"
// ============================================================
// ADAFRUIT IO SETTINGS
// ============================================================
#define AIO_USERNAME    "Hari_87789"
#define AIO_KEY         "aio_mTwj83UKedBOZF7ynQuOfANn9JK0"
#define AIO_SERVER      "io.adafruit.com"
#define AIO_SERVERPORT  1883
// ============================================================
// PIN CONFIGURATION
// ============================================================
// PIR Sensor
#define PIR_PIN          14      // D5 / GPIO14
// Servo Motor
#define SERVO_PIN        12      // D6 / GPIO12
// Active-LOW Buzzer
#define BUZZER_PIN        2      // D4 / GPIO2
// Green LED
#define GREEN_LED        13      // D7 / GPIO13
// Red LED
```

```
#define RED_LED           0     // D3 / GPIO0
// UP Push Button
#define UP_SWITCH         5     // D1 / GPIO5
// DOWN Push Button
#define DOWN_SWITCH       4     // D2 / GPIO4
// ========================================================
// SERVO SETTINGS
// ========================================================
#define DOOR_CLOSED       0
#define DOOR_OPEN         180
// ========================================================
// OBJECTS
// ========================================================
Servo elevatorServo;
WiFiClient client;
Adafruit_MQTT_Client mqtt(
  &client,
  AIO_SERVER,
  AIO_SERVERPORT,
  AIO_USERNAME,
  AIO_KEY
);
// ========================================================
// ADAFRUIT IO FEEDS
// ========================================================
// Human detection feed
Adafruit_MQTT_Publish humanFeed =
  Adafruit_MQTT_Publish(
    &mqtt,
    AIO_USERNAME "/feeds/human"
```

```
  );
// Elevator movement/status feed
Adafruit_MQTT_Publish movementFeed =
  Adafruit_MQTT_Publish(
    &mqtt,
    AIO_USERNAME "/feeds/movement"
  );
// =============================================================
// VARIABLES
// =============================================================
bool humanDetected = false;
bool elevatorBusy = false;
bool lastUpState = HIGH;
bool lastDownState = HIGH;
// =============================================================
// WIFI CONNECTION
// =============================================================
void connectWiFi()
{
  Serial.println();
  Serial.println("Connecting to WiFi...");
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  Serial.println();
  Serial.println("WiFi Connected!");
```

```
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}
// =====================================================
// ADAFRUIT IO CONNECTION
// =====================================================
void connectMQTT()
{
  if (mqtt.connected())
  {
    return;
  }
  Serial.println("Connecting to Adafruit IO...");
  int8_t ret;
  while ((ret = mqtt.connect()) != 0)
  {
    Serial.println(mqtt.connectErrorString(ret));
    mqtt.disconnect();
    delay(5000);
    Serial.println("Retrying Adafruit IO...");
  }
  Serial.println("Adafruit IO Connected!");
}
// =====================================================
// SEND HUMAN STATUS
// =====================================================
void sendHuman(const char *status)
{
  Serial.print("HUMAN: ");
  Serial.println(status);
```

```
  if (!humanFeed.publish(status))
  {
    Serial.println("Failed to publish human status!");
  }
}
// ========================================================
// SEND MOVEMENT STATUS
// ========================================================
void sendMovement(const char *status)
{
  Serial.print("MOVEMENT: ");
  Serial.println(status);
  if (!movementFeed.publish(status))
  {
    Serial.println("Failed to publish movement status!");
  }
}
// ========================================================
// BUZZER OFF
// Active LOW buzzer:
// LOW  = ON
// HIGH = OFF
// ========================================================
void buzzerOff()
{
  digitalWrite(BUZZER_PIN, HIGH);
}
// ========================================================
// SHORT BEEP
// ========================================================
```

```
void shortBeep()
{
  // Buzzer ON
  digitalWrite(BUZZER_PIN, LOW);
  delay(200);
  // Buzzer OFF
  digitalWrite(BUZZER_PIN, HIGH);
}
// ========================================================
// DOOR OPEN
// Servo = 180°
// Slow beep for 5 seconds
// ========================================================
void openDoor()
{
  Serial.println("DOOR OPEN");
  sendMovement("DOOR OPEN");
  // ------------------------------------------
  // OPEN SERVO
  // ------------------------------------------
  elevatorServo.write(DOOR_OPEN);
  // ------------------------------------------
  // LED STATUS
  // ------------------------------------------
  digitalWrite(GREEN_LED, HIGH);
  digitalWrite(RED_LED, LOW);
  // ------------------------------------------
  // SLOW BUZZER FOR 5 SECONDS
  // ------------------------------------------
  Serial.println("Door open - warning buzzer");
```

```
  unsigned long startTime = millis();
  while (millis() - startTime < 5000)
  {
    // Buzzer ON
    digitalWrite(BUZZER_PIN, LOW);
    delay(200);
    // Buzzer OFF
    digitalWrite(BUZZER_PIN, HIGH);
    delay(800);
    // Keep Adafruit IO alive
    if (mqtt.connected())
    {
      mqtt.processPackets(10);
    }
  }
  // IMPORTANT:
  // Buzzer must be OFF after door-open period
  digitalWrite(BUZZER_PIN, HIGH);
  Serial.println("Door warning buzzer OFF");
}
// ========================================================
// DOOR CLOSE
// Servo = 0°
// Buzzer OFF
// ========================================================
void closeDoor()
{
  Serial.println("DOOR CLOSE");

  sendMovement("DOOR CLOSE");
```

```
  // ---------------------------------------------
  // STOP BUZZER IMMEDIATELY
  // ---------------------------------------------
  digitalWrite(BUZZER_PIN, HIGH);
  // ---------------------------------------------
  // CLOSE SERVO
  // ---------------------------------------------
  elevatorServo.write(DOOR_CLOSED);
  // ---------------------------------------------
  // LED STATUS
  // ---------------------------------------------
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(RED_LED, HIGH);
  delay(1000);
  // Make sure buzzer remains OFF
  digitalWrite(BUZZER_PIN, HIGH);
}
// ================================================
// ELEVATOR UP OPERATION
// ================================================
void elevatorUp()
{
  elevatorBusy = true;
  Serial.println();
  Serial.println("==================================");
  Serial.println("       ELEVATOR UP");
  Serial.println("==================================");
  // ================================================
  // 1. DOOR OPEN
  // ================================================
```

```
openDoor();
// ====================================================
// 2. DOOR CLOSE
// ====================================================
closeDoor();
// ====================================================
// 3. MOVING UP
// ====================================================
Serial.println("MOVING UP");
sendMovement("MOVING UP");
digitalWrite(GREEN_LED, HIGH);
digitalWrite(RED_LED, LOW);
// Buzzer must be OFF
buzzerOff();
// Simulate elevator movement
delay(5000);
// ====================================================
// 4. STOPPED
// ====================================================
Serial.println("STOPPED");
sendMovement("STOPPED");
digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, HIGH);
// Buzzer OFF
buzzerOff();
delay(1000);
// ====================================================
// 5. DESTINATION DOOR OPEN
// ====================================================
openDoor();
```

```
  // ================================================
  // 6. DESTINATION DOOR CLOSE
  // ================================================
  closeDoor();
  // ================================================
  // 7. IDLE
  // ================================================
  Serial.println("IDLE");
  sendMovement("IDLE");
  elevatorBusy = false;
  Serial.println("UP operation completed.");
  Serial.println("Waiting for next command...");
}
// ========================================================
// ELEVATOR DOWN OPERATION
// ========================================================
void elevatorDown()
{
  elevatorBusy = true;
  Serial.println();
  Serial.println("==================================");
  Serial.println("      ELEVATOR DOWN");
  Serial.println("==================================");
  // ================================================
  // 1. DOOR OPEN
  // ================================================
  openDoor();
  // ================================================
  // 2. DOOR CLOSE
  // ================================================
```

```
closeDoor();
// ====================================================
// 3. MOVING DOWN
// ====================================================
Serial.println("MOVING DOWN");
sendMovement("MOVING DOWN");
digitalWrite(GREEN_LED, HIGH);
digitalWrite(RED_LED, LOW);
// Buzzer OFF
buzzerOff();
// Simulate elevator movement
delay(5000);
// ====================================================
// 4. REACHED
// ====================================================
Serial.println("REACHED");
sendMovement("REACHED");
digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, HIGH);
// Buzzer OFF
buzzerOff();
delay(1000);
// ====================================================
// 5. DESTINATION DOOR OPEN
// ====================================================
openDoor();
// ====================================================
// 6. DESTINATION DOOR CLOSE
// ====================================================
closeDoor();
```

```
  // ======================================================
  // 7. IDLE
  // ======================================================
  Serial.println("IDLE");
  sendMovement("IDLE");
  elevatorBusy = false;
  Serial.println("DOWN operation completed.");
  Serial.println("Waiting for next command...");
}
// ======================================================
// SETUP
// ======================================================
void setup()
{
  Serial.begin(115200);
  delay(1000);
  Serial.println();
  Serial.println("========================================");
  Serial.println("     SMART ELEVATOR AUTOMATION");
  Serial.println("========================================");
  // ======================================================
  // PIN MODES
  // ======================================================
  pinMode(PIR_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  // Push buttons
  pinMode(UP_SWITCH, INPUT_PULLUP);
  pinMode(DOWN_SWITCH, INPUT_PULLUP);
```

```
// ================================================
// SERVO
// ================================================
elevatorServo.attach(SERVO_PIN);
// Door initially closed
elevatorServo.write(DOOR_CLOSED);
// ================================================
// INITIAL OUTPUTS
// ================================================
// Active-LOW buzzer:
// HIGH = OFF
digitalWrite(BUZZER_PIN, HIGH);
// Green OFF
digitalWrite(GREEN_LED, LOW);
// Red ON
digitalWrite(RED_LED, HIGH);
// ================================================
// WIFI
// ================================================
connectWiFi();
// ================================================
// ADAFRUIT IO
// ================================================
connectMQTT();
// ================================================
// INITIAL CLOUD STATUS
// ================================================
sendHuman("NO HUMAN");
sendMovement("IDLE");
```

```
  Serial.println();
  Serial.println("--------------------------------------");
  Serial.println("SYSTEM READY");
  Serial.println("Door     : CLOSED");
  Serial.println("Elevator : IDLE");
  Serial.println("Human    : NO HUMAN");
  Serial.println("Buzzer   : OFF");
  Serial.println("Waiting for human...");
  Serial.println("--------------------------------------");
}
// ====================================================
// MAIN LOOP
// ====================================================
void loop()
{
  // ====================================================
  // WIFI
  // ====================================================
  if (WiFi.status() != WL_CONNECTED)
  {
    connectWiFi();
  }
  // ====================================================
  // ADAFRUIT IO
  // ====================================================
  if (!mqtt.connected())
  {
    connectMQTT();
  }
  mqtt.processPackets(10);
```

```
mqtt.ping();
// ====================================================
// PIR HUMAN DETECTION
// ====================================================
int pirState = digitalRead(PIR_PIN);
// ====================================================
// HUMAN DETECTED
// ====================================================
if (pirState == HIGH && !humanDetected)
{
  humanDetected = true;
  Serial.println();
  Serial.println(">>> HUMAN DETECTED <<<");
  sendHuman("HUMAN DETECTED");
  // Short notification beep
  shortBeep();
  delay(200);
}
// ====================================================
// NO HUMAN
// ====================================================
if (pirState == LOW && humanDetected && !elevatorBusy)
{
  humanDetected = false;
  Serial.println(">>> NO HUMAN <<<");
  sendHuman("NO HUMAN");
}
// ====================================================
// BUTTON CONTROL
// ONLY WHEN HUMAN IS DETECTED
```

```
// ================================================
if (humanDetected && !elevatorBusy)
{
 int upState = digitalRead(UP_SWITCH);
 int downState = digitalRead(DOWN_SWITCH);
 // ================================================
 // UP BUTTON
 // Detect HIGH → LOW
 // ================================================
 if (lastUpState == HIGH && upState == LOW)
 {
  delay(50);
  // Debounce
  if (digitalRead(UP_SWITCH) == LOW)
  {
   Serial.println();
   Serial.println(">>> UP BUTTON PRESSED <<<");
   shortBeep();
   elevatorUp();
  }
 }
 // ================================================
 // DOWN BUTTON
 // Detect HIGH → LOW
 // ================================================
 if (lastDownState == HIGH && downState == LOW)
 {
  delay(50);
  // Debounce
  if (digitalRead(DOWN_SWITCH) == LOW)
```

```
    {
      Serial.println();
      Serial.println(">>> DOWN BUTTON PRESSED <<<");
      shortBeep();
      elevatorDown();
    }
  }
  // Save current button states
  lastUpState = upState;
  lastDownState = downState;
}
else
{
  // Update button states
  lastUpState = digitalRead(UP_SWITCH);
  lastDownState = digitalRead(DOWN_SWITCH);
}
// Small loop delay
delay(50);
```

System Working:

- The system starts by initializing the ESP32/Arduino, PIR sensor, floor buttons, motor driver, door motor, and display. The PIR sensor detects human presence and sends the signal to the microcontroller. When a person is detected, the elevator door opens automatically, allowing the user to enter and select the desired floor. The controller then checks the door safety condition before closing it.
- If a person is detected while the door is closing, the door immediately stops and reopens to prevent injury. Once the door is safely closed, the elevator motor moves the cabin to the selected floor.

- When the destination floor is reached, the motor stops and the floor number is displayed. The door then opens automatically to allow the passenger to exit. After detecting that the passenger has left, the door closes and the elevator returns to **idle mode**, ready for the next operation.

Bill Of Material:

S. no	Component	Specification	Quantity	Cost
1	ESP8266	Node MCU ESP8266	1	Rs. 450
2	PIR Sensor	HC-SR501	1	Rs. ₹80
3	Servo Motor	SG90	1	Rs. 100
4	Bread Board	Standard 830-point	1	Rs. 100
5	Jumper Wire	Male-Male / Male-Female	1 set	₹50
		Total		₹780

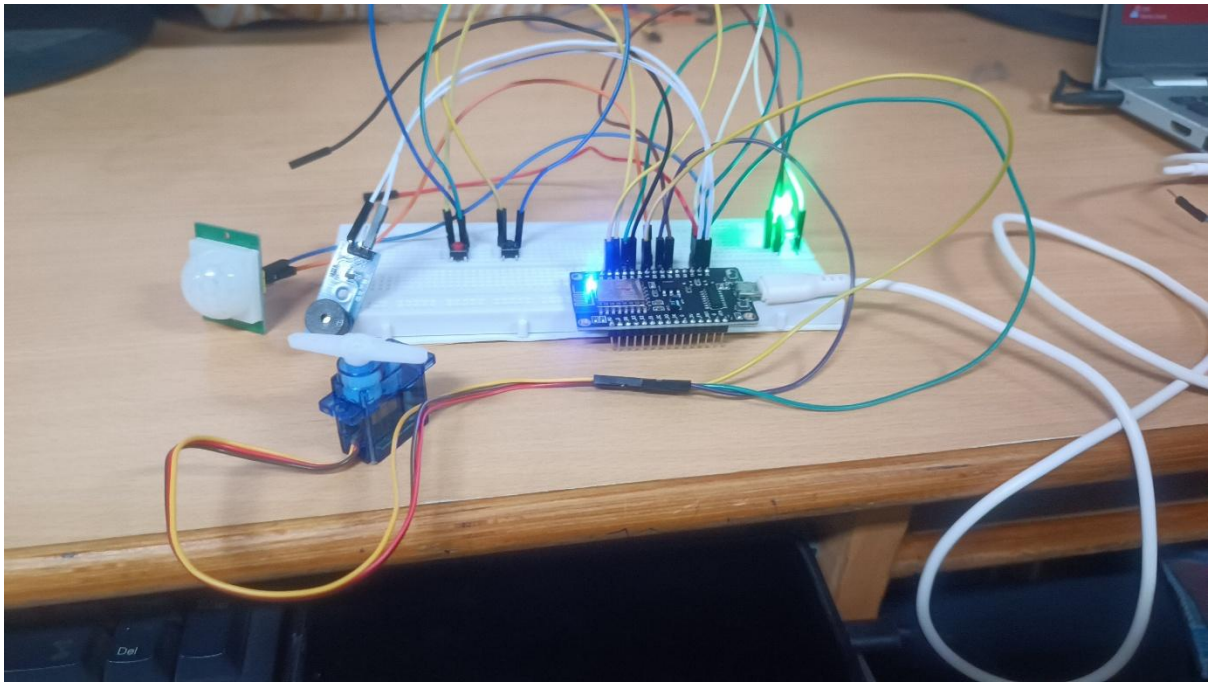
Prototype Implementation:

- The prototype was implemented using an ESP32 microcontroller, PIR sensor, servo motor, push buttons, OLED display, LEDs and buzzer. The PIR sensor is positioned near the elevator entrance to detect the presence of a person.
- When human presence is detected, the ESP32 processes the PIR sensor signal and activates the elevator system. The user can select the required floor using the push buttons, and the selected floor is displayed on the OLED screen. The servo motor is used to simulate the opening and closing of the elevator door.
- The LEDs indicate the operating status of the elevator, while the buzzer provides an alert during safety conditions or when human presence is detected. The ESP32 coordinates the sensor inputs and elevator operations to provide automated and safe functioning.

The prototype follows the data flow:

PIR Sensor → ESP32 → Human Detection → Floor Selection → Elevator Operation → Servo Door Control → OLED + LED/Buzzer

The implemented prototype demonstrates a low-cost and automated elevator safety system that can improve human detection, floor selection and elevator operation.



Results and Performance:

- The developed Smart Elevator Automation and Human Detection System successfully detected human presence using the PIR sensor and controlled the elevator operations through the ESP32. The system responded to human movement and initiated the elevator operation when a person was detected.
- The selected floor was entered using the push buttons and displayed on the OLED screen. The servo motor successfully simulated elevator door opening and closing, while the LEDs provided visual status indications. The buzzer generated an alert during specified safety conditions.

Performance Analysis

Parameter	Result
Human Detection	Successfully detected using PIR sensor
Floor Selection	Successfully controlled using push buttons
Elevator Status	Displayed on OLED
Door Operation	Simulated using servo motor

Safety Alert	Buzzer activated when required
Visual Indication	LEDs provided status indication
Response	Real-time
Automation	Successfully implemented

Overall, the prototype demonstrated a low-cost, responsive and automated elevator control system. The PIR-based human detection improves safety by ensuring that elevator operations respond to the presence of users. The system can be further enhanced by adding multiple floor sensors, ultrasonic obstacle detection, motor-based elevator movement, overload detection and emergency-stop functionality.